# 2020 AMC 10A Problem 18 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

## 2020 AMC 10A Problem 18

**Problem:**
Let $(a, b, c, d)$ be an ordered quadruple of not necessarily distinct integers, each one of them in the set $\{0, 1, 2, 3\}$. For how many such quadruples is it true that $a \cdot d - b \cdot c$ is odd? (For example, $(0, 3, 1, 1)$ is one such quadruple, because $0 \cdot 1 - 3 \cdot 1 = -3$ is odd.)

**Answer Choices:**

A. $48$
B. $64$
C. $96$
D. $128$
E. $192$

**Solution:**

Consider parity. We need exactly one term to be odd, one term to be even. Because of symmetry, we can set $ad$ to be odd and $bc$ to be even, then multiply by $2$.

If $ad$ is odd, both $a$ and $d$ must be odd, therefore there are

$$2 \cdot 2 = 4 \text{ possibilities for } ad$$

Consider $bc$. Let us say that $b$ is even. Then there are

$$2 \cdot 4 = 8 \text{ possibilities for } bc$$

However, $b$ can be odd, in which case we have

$$2 \cdot 2 = 4 \text{ more possibilities for } bc$$

Thus there are $12$ ways for us to choose $bc$ and 4 ways for us to choose $ad$. Therefore, also considering symmetry, we have

$$2 \cdot 4 \cdot 12 = \boxed{(C)\ 96} \text{ total values of } ad - bc$$

$$\textbf{OR}$$

There are $4$ ways to choose any number independently and $2$ ways to choose any odd number independently. To get an even products, we count:

$$\text{P( any number )} \cdot \text{P( any number )} - \text{P( odd )} \cdot \text{P( odd )} = 4 \cdot 4 - 2 \cdot 2 = 12$$

The number of ways to get an odd product can be counted like so:

$$\text{P( odd )} \cdot \text{P( odd )} = 2 \cdot 2 = 4$$

So, for one product to be odd the other to be even:

$$2 \cdot 4 \cdot 12 = \boxed{(C)\ 96} \text{ (order matters)}$$